PROJECT-BASED LEARNING

PHASE-I EVALUATION

RakshaNetra - NHAI Worker Safety and Emergency Response System

Team ID: DSCPP-III-2026-T072

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T072
Semester: 3rd
Section: DS1,AIML1,AIML2,AIML3
Domain: *AIML and AIDS*

Team Members

1. *Soumya Rawat – 2510380027*
2. *Tazeen Anjum – 2510370373*
3. *Sarthak Singh Chauhan– 2510012694*
4. *Pranjal Pandey – 2510370507*

Mentor

Dr. Amit Kumar **Interactions completed:** ______

Problem Statement & Motivation

Problem Statement

NHAI construction sites involve many workers, staff, vehicles, and locations, making safety records difficult to manage manually. During emergencies, finding worker details and arranging suitable rescue support can take valuable time.The project aims to provide a simple system for worker management, identity verification, and faster emergency response.

Motivation

- *Manual and scattered records.*
- *Delayed emergency response.*
- *No single safety system.*

Target Users / Stakeholders

- *NHAI site staff and workers.*
- *Rescue teams and site managers.*
- *Better safety and faster emergency response.*

Objectives

Project Objectives

1. *Develop a simple and organized system to manage worker safety and emergency situations at NHAI construction sites.*
2. *Manage worker records, verification, safety, and emergencies.*
3. *Integrate concepts from both PBL subjects by using Data Structures in C and Object-Oriented Programming in C++.*
4. *Test the system with different worker, safety, and emergency situations.*
5. *Demonstrate the practical usefulness of the system and its future scope.*

Key Objective: *Improve NHAI worker safety through quick identification and emergency response.*

Proposed Solution

Proposed Approach

1. *Collect worker, staff, site, and emergency details.*
2. *Verify workers and assess emergencies.*
3. *Prioritize emergencies and assign rescue teams.*
4. *Store all records systematically.*
5. *Display required safety and emergency information.*

Key Features

- *Worker and staff record management*
- *Worker identity verification*
- *Safety incident recording*
- *Emergency severity and priority management*
- *Rescue team selection and assignment*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, Linked Lists, Queues, Searching, Sorting*	*Worker and Emergency Management*
OOPs with C++	*Classes, Objects, Inheritance, Polymorphism*	*Rescue Team Management*

Requirement: Data Structures and OOPs concepts will work together to manage worker records, handle emergencies, and support faster rescue response.

System Architecture / Workflow

Worker and Emergency Details → Data validation → Verification and emergency handling → Store records → Safety status

Major Components

- *Worker Management*
- *Safety and Emergency Management*
- *Rescue Team Management*

Data / Control Flow

- *Worker data → Processing → Database → Output*
- *Modules work together for emergency response.*
- *Worker,emergency,rescue details.*

Technology Stack & Methodology

Technology Stack

- Programming: *C and C++*
- Database: *MySQL*
- Tools / Framework: *STL*
- IDE: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C and C++ support Data Structures and OOP concepts, while MySQL helps store data in an organized way.VS Code, GitHub, and the step-by-step methodology make development and testing easier.

Initial Progress & Team Contribution

Progress Achieved

- *NHAI worker safety issues identified.*
- *Worker and emergency requirements identified.*
- *C/C++ system architecture and complete workflow prepared.*
- *Initial worker and emergency modules implemented.*
- *Basic data structures and data storage prepared.*

Individual Contribution

Member	Role	Contribution
Soumya Rawat	*Lead*	*Planning and coordination*
Sarthak Singh Chauhan	*Developer*	*C/C++ module development*
Pranjal Pandey	*DB/Testing*	*Database and testing*
Tazeen Anjum	*Developer/Docs*	*Development and documentation*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Working safety system*
- *Faster emergency response*
- *Better worker management*
- *Can be expanded with real-time tracking and alerts*

References

1. *NHAI Official Website*
2. *Road Safety Research Papers*
3. *C/C++ Documentation*
4. *MySQL Documentation*

Thank You
Questions & Suggestions